

Cattle Infectious Disease Prediction Using Machine Learning Model and Explainable AI

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Abstract— Cattle infectious disease prediction can save cattle's life and increase productivity. The existing cattle disease prediction works did not work on multi class cattle disease prediction and explainable AI based interpretation. This paper proposes a machine learning-based multiclass cattle illness prediction system to address these issues. This paper employed a large dataset that included important characteristics like species, age, temperature, immunization status, and symptoms, as well as a variety of illness types. The proposed method employed SHAP analysis to determine the most useful variables, Recursive Feature Addition (RFA) based feature selection, and SMOTENC oversampling to solve class imbalance. GridSearchCV and RandomizedSearchCV were used to train and hyper tune a number of ML models. According to the results, the proposed XGBoost method outperformed previous works by at least 1% and up to 12% accuracy and F1-score gain.

Keywords— Cattle disease prediction, machine learning, SHAP, RFA, SMOTE-NC, XGBoost.

I. INTRODUCTION

The cattle industry is essential to global agriculture because it gives people and communities around the world jobs, income, and nutrition. According to the FAO, around 1.3 billion people worldwide depend on cattle for their livelihoods and health. Furthermore, it contributes 40% of the world's agricultural production [1]. The livestock sector in Bangladesh is growing at a pace of 3.47% and contributes 1.47% of the country's GDP. About 20% of people are directly employed by the livestock sector [2]. With over 24.7 million cattle and 1.5 million buffalo [3], infectious illnesses pose a constant threat to Bangladesh's livestock sector, resulting in catastrophic financial losses from cow deaths and undermining rural livelihoods. Due to the death of cattle's regarding foot and mouth disease, almost 2 billion money is lost [4]. Since 2019, Lumpy Skin Disease (LSD) has been a major hazard, causing an estimated BDT 8,203.22 crore in damage nationwide [5], [6]. In just two areas, bovine tuberculosis causes an additional BDT 1,347.72 million in damages each year [7]. These illnesses have a domino impact on the agricultural value chain because they lower milk output, lower the quality of meat, raise veterinary expenses, and impair reproductive function [8]. In Bangladesh's livestock industry, the traditional approaches to disease diagnosis mostly involve physical

examinations, subjective clinical indicator-based questions, and syndrome- or symptom-oriented assessments. They take a lot of time and call for high levels of veterinary expertise, which are frequently lacking or unavailable in the countryside. Clinical diagnoses typically involve comprehensive assessments, physical examinations, and simple microscopic examinations carried out using standard laboratory procedures; nonetheless, they are subject to a number of limitations [9]. Long wait times, lack of access to cutting-edge diagnostic tools, and the time and financial costs of traveling to offer veterinary services are other significant obstacles faced by cattle farmers [10], [11].

Machine learning (ML) has shown promise in early cattle illness identification, according to [12], [7], [13], and [14]. Research has demonstrated that ML algorithms are capable of efficiently examining animal traits and symptom patterns to forecast illnesses before they spread widely. Only binary classification is used in the existing efforts on the prediction of infectious diseases in cattle. For the prediction of multiple classes of cattle infectious diseases, such as Lumpy Skin Disease, Mastitis, Anthrax, Bovine Brucellosis, Hemorrhagic Septicemia, foot and mouth issue, Blackleg, and Healthy class, the prior works did not suggest any ML based prediction system with high accuracy. Class imbalance issues, poor prediction accuracy, improper feature extraction, improper use of explainable AI methods (EAI), and a lack of hyper parameter tweaking strategies were all unresolved by existing works. In order to overcome the constraints, this paper presents a ML based model for predicting infectious diseases and healthy cases in cattle. The contribution lists of this article are given as follows:

- (i) By incorporating a wider variety of cow diseases, symptoms, this work gathered an extensive dataset for the prediction of infectious diseases in cattle.
- (ii) In this paper, several infectious diseases in cattle were predicted using various ML models. The efficient ML model was selected for prediction with its high accuracy.
- (iii) By combining SMOTE-NC for oversampling, Recursive Feature Addition (RFA) for feature selection, and extensive hyper parameter tuning, the proposed XGBoost model achieved better performance metrics than earlier works. The SHAP based Explainable AI technique was used in this paper to discuss features impact on models output.

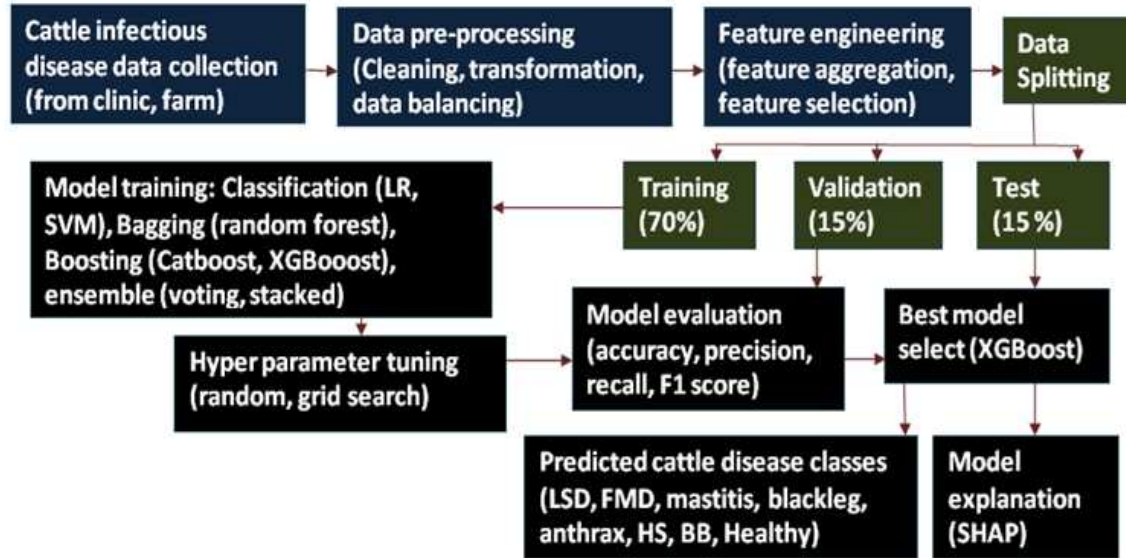


Fig. 1. Proposed methodology for cattle infectious disease prediction

The related works are covered in section II. The section III describes the proposed system methodology. The evaluation results are provided in section IV. The main conclusion of this paper is outlined in section V.

II. LITERATURE REVIEW

The literary works associated with cattle infectious disease prediction is discussed in this section. A Cat Boost-based cattle illness prediction technique utilizing characteristics like age, symptoms, temperature, and disease kind was given in [12]. The lack of oversampling to solve class imbalance, poor accuracy (83.72%) in comparison to other approaches, and the lack of explainable AI methodologies, were its limitations.

Using geospatial and meteorological data, the authors of [15] used Random Forest for LSD disease prediction. They suffered from over fitting problems and binary classification problems. Using geographic and climatic characteristics, the study in [16] used SVM for LSD disease prediction due to their high accuracy. Using TF-IDF for feature extraction, the study in [17] investigated the performance of ML models for the bovine diseases prediction using symptom-based text data. According to their research, CNN and XGBoost performed better when handling diseases with overlapping or unclear symptom patterns, whereas MLP performed exceptionally well when recognizing illnesses with unique and obvious symptoms, such anthrax and blackleg. They failed to use appropriate feature engineering techniques and did not look into the issue of class imbalance. By combining manual health examinations with ML and precise IoT technologies like sensors and automatic feeders, the study in [18] predicted bovine diseases or BRD. In addition to achieving up to 88% accuracy for BRD diagnosis, their method introduced the CALF feature selection method.

The study in [19] used a time-series based machine learning strategy to predict subclinical mastitis in dairy cows. The main limitations of their approach are low accuracy value (e.g., 71 percent). The study in [20] investigated the DL technique for cattle's FMD disease detection using image processing techniques. However, additional cattle diseases are not covered by the study because it only addressed binary classification (healthy vs. FMD). Using a single accelerometer on the left flank of the cow, the work in [13] used ML algorithms for cow's behavior recognition. However, the system's overall behavior recognition accuracy was below 80%. A DL-based method for predicting skin diseases in cattle was developed in [21]. They did not, however, address the problem of numerous infectious cow illnesses. The influence of CNN model employment for lumpy skin disease prediction was examined by the authors in [22]. The influence of IoT devices, sensors, and GPS technology for real-time cow illness diagnosis was examined in the paper in [23]. A DL-based foot and mouth disease prediction system for cattle was created in [24]. Fuzzy maps and genetic algorithms were both used in the work in [25] to generate treatment recommendations for cow diseases.

The article in [26] used an IoT sensor and a Raspberry Pi to create a real-time cow wellness monitoring system. A DL-based Brucellosis outbreak prediction system for dairy cattle was created in [27]. A symptom-based multiple cattle infectious disease prediction system utilizing ML and explainable AI was not developed by earlier works. The current efforts lack solutions for resolving class imbalance issues, feature engineering, and tuning of parameter. This work presents a multi-class cattle disease prediction system based on symptoms that uses ML and explainable AI with high accuracy, appropriate feature engineering, handling of class imbalances, and tuning of parameter.

III. PROPOSED SCHEME

This section contains the system model regarding ML based cattle infectious disease prediction.

A. System Model

Figure 1 displays the proposed methodology for cattle infectious disease prediction. Every step required for prediction is covered by the methodology, including feature engineering, dataset splitting, data collecting, preprocessing, model training, hyper parameter tuning, and model evaluation. Data for this study was gathered from farms and livestock clinics. This data preprocessing process includes data transformation, data cleaning, and data balancing. Using various techniques, this paper completed the feature engineering's feature selection and feature aggregation tasks. For the purpose of predicting the multiple infectious diseases in cattle, this paper investigated various single machine learning models (LR, XGBoost, and RF), bagging, boosting, and ensemble models. The best ML model was chosen by assessing a number of performance criteria, such as accuracy. This paper used random search along with grid search for hyper parameter tuning. This paper explained the interpretability of the models using an analysis based on SHAP.

B. Dataset Preparation

The dataset was gathered from various Chittagong livestock farms. The veterinary physicians in Bangladesh supplied the questionnaire that was used to generate the dataset, and the veterinary specialists in Bangladesh also verified the dataset. Dataset was generated by asking several questions relating to cattle's disease symptoms to the cattle owners and farmers. Figure 2 displays a sample of the cow illness dataset with symptoms. There are nine column entries and 1910 rows in the dataset. Three categories of data are included in the dataset: categorical (animal, disease, vaccination, organization, location), numerical (age, temperature), and list of strings (symptoms).

The dataset's labeled data includes one healthy class and seven diseases. The following classes have been labeled: Blackleg, Hemorrhagic Septicemia, Mastitis, Anthrax, FMD disease, LSD (lumpy) disease, Bovine Brucellosis, and Healthy class.

C. Data cleaning and preprocessing

Data cleaning, transformation (scaling, encoding), and data set balancing are the three components of data preprocessing steps (e.g., data cleaning). First, this work looked for any missing or null values. Suppressing null or empty fields in the dataset is a preprocessing step. The Interquartile Range (IQR) method was then used to check for outliers in the Age and Temperature data. It was discovered that both variables fell within the expected ranges and that no data points were found to be outliers. To guarantee data quality, the dataset was examined for duplicates, and two identical rows were identified and eliminated. Data transformation comes next. In order to standardize variance, StandardScaler was used to numerical features such as temperature and age. Three distinct encoding methods were used in this work to change the category data. The symptoms column, which has lists of several categorical values, was first encoded using the MultiLabelBinarizer. For every distinct symptom, this method produced binary columns that indicated whether the

symptom was present or not. Since the category "vaccinated" has a greater priority than "not vaccinated," an ordinal relationship is acceptable; hence this paper used the Ordinal Encoder for the Vaccination column. The target labels were then encoded using the Label Encoder, which assigned integer values ranging from 0 to 7 to represent the eight different disease groups. The example values following the use of encoding techniques are displayed in Figure 3(a) and 3(b).

This study used SMOTE-NC technique, which is designed especially for datasets with both categorical features and numerical features, to correct class imbalances in the dataset. In contrast to standard SMOTE, SMOTE-NC replicates the most frequent categorical feature from neighbors and interpolates numerical characteristics to create synthetic samples using K-nearest neighbors. In datasets pertaining to healthcare, like as this one, it is particularly helpful because categorical variables—such species type, symptoms, and vaccination status—are just as significant as numerical ones, like age and temperature. This study reduced bias towards majority classes, provided appropriate balance among all sickness classes using SMOTE-NC. The label class counts after samplings (SMOTE NC apply) are displayed in Figure 3(c). The number of samples for anthrax, FMD, hemorrhagic septicemia (HS), mastitis (mas), bovine brucellosis (Bb), lumpy skin (LSD), health (he), and blackleg (Bl) are before SMOTE-NC apply are 168, 167, 167, 172, 168, 173, 176, and 167, respectively. After the SMOTE-NC applies, the record under each class is 176.

D. Feature Selection, Dataset splitting

To make sure the model concentrated on the most pertinent predictors for precise and understandable cattle disease diagnosis, feature selection techniques like Recursive Feature addition were then applied. In order to measure the significance of each feature in predicting cattle diseases, our work started the feature engineering process by training a classifier on our preprocessed dataset. While symptoms like sneezing, a faint cough, and a slight limp contributed less to the model's predictions, the study showed that the most important features were temperature, age, fever, lameness, and decreased milk supply.

Figure 4 illustrates the importance of features. In this work, features with an importance of less than two were grouped into semantically relevant categories. The binary indicators of the mentioned symptoms were added up in this article, and the result was clipped to a maximum of 1. After aggregating the dataset, this approach eliminated the original individual symptom features to prevent redundancy. This study first iteratively tried multiple feature counts, monitoring macro F1 scores on the validation set for each in order to identify the most informative features. In order to determine the most significant features, this work used Recursive Feature Addition with Random Forest on the training data after discovering that 30 out of 36 features provided the optimum balance between performance and complexity. Majority of the data was used for training purpose (70%). The 15 percent data is for test purpose. The remaining data is applied for validation purpose.

Age	Temperature (degree F)	Vaccine	Symptoms
1	104.8 cow	No	Salivation, no swelling, skin nodules, blisters, fever, swollen abdomen, lameness
4	102 buffalo	No	Salivation, no swelling, udder swelling, blisters, fever, swollen abdomen, lameness
2	104.3 buffalo	No	Salivation, blisters, fever, alert, lameness, respiratory distress
2	104.4 cow	No	Salivation, nasal discharge, muscle stiffness, blisters, fever, swollen abdomen, lameness

(a)

Disease	Collected from	Organization	Location
Foot and mouth disease	Dr. S. Parvez	Upazila livestock veterinary hospital	Chandpur
Foot and mouth disease	Nizam Mia	Shah amanat agro	Chittagong
Foot and mouth disease	Dr. S. Parvez	Upazila livestock veterinary hospital	Chandpur
Foot and mouth disease	Dr. S. Parvez	Upazila livestock veterinary hospital	Chandpur

(b)

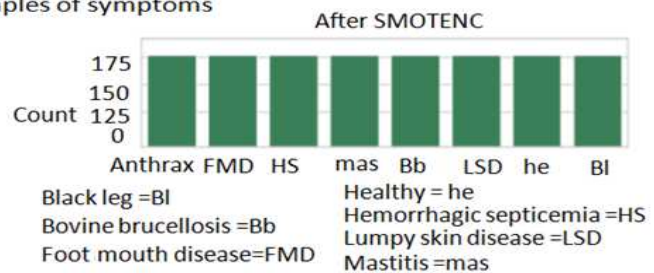
Fig. 2. Glimpse of the dataset

Abortion	Alert	Anemia	Blisters	Bloody discharge	Dry skin	Ear twitching
1	0	0	0	1	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	0	0	0	0
0	1	1	0	0	0	0

(a) Partial binarized encoded samples of symptoms

Encoded	Original label
0	Anthrax
1	Black leg
2	Bovine brucellosis
3	Foot and mouth disease
4	healthy
5	Hemorrhagic septicemia
6	Lumpy skin disease
7	mastitis

(b) Label encoded samples of disease



(c)

Fig. 3. (a) Binarized encoded samples, (b) Label encoding, (c) SMOTENC apply for class balancing

Feature	importance	feature	Importance	feature	importance
Temp.	14.68	blisters	3.185	Swollen limb	2.640
age	9.25	Appetite	3.183	vaccination	2.618
fever	7.77	Muscle stiffness	3.106	infertility	2.000
lameness	6.33	Nasal discharge	3.075	Joint swelling	1.843
Reduced milk yield	5.31	Bloody discharge	2.93	abortion	1.805
Udder swelling	3.53	Skin nodules	2.906	Swollen abdomen	1.776
salivation	3.32	Gas under skin	2.750	anemia	1.524
Res. distress	3.26	Sudden death	2.703	No swelling	1.455
alert	1.385	Mild cough	0.502	swelling	.383
Good coat	1.168	Slight limp	.471	Tail biting	.368
sneezing	0.66	Ear twitching	.444		

Fig. 4. Selected features list and feature importance ranking

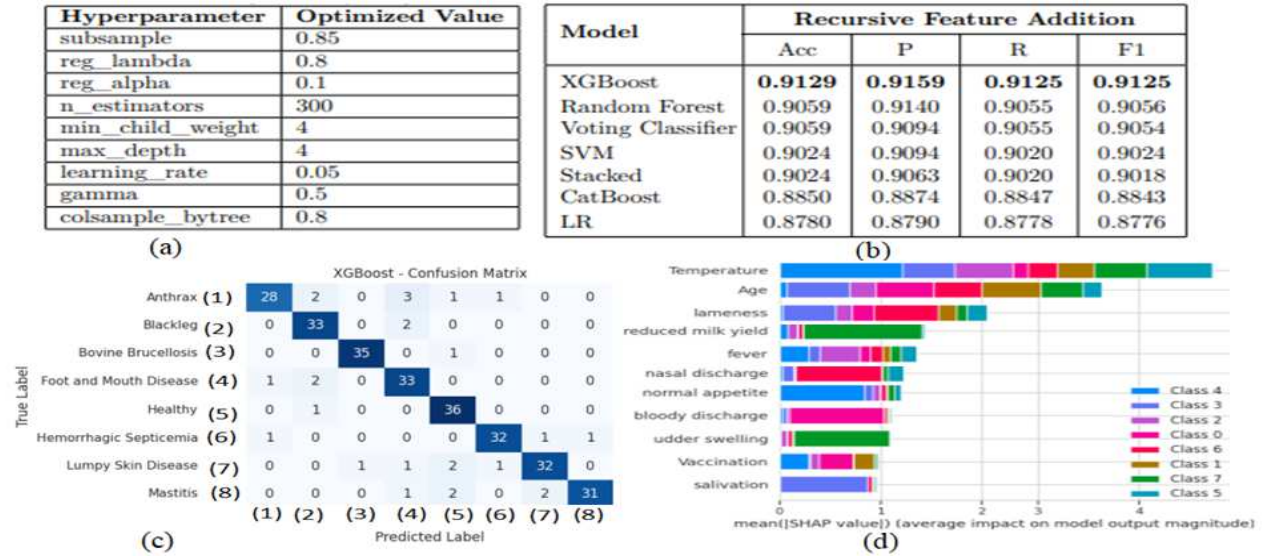


Fig. 5. (a) Hyper parameter settings for XGBoost, (b) ML model performance results, (c) Confusion matrix (XGBoost), (d) SHAP summary plot

TABLE I
COMPARISON RESULT

Work	Precision	Recall	F1 score	Accuracy
Utami et al. [15], AdaBoost	91.26%	90.59%	90.58%	90.59%
Bhardwaj et al. [12], CatBoost	88.94%	88.53%	88.53%	88.48%
Naik et al.,[17], MLP	81.84%	81.15%	81.18%	81.15%
Our Proposed Method, XGBoost	91.59%	91.25%	91.25%	91.29%

E. Model Selection, Hyper-parameter Tuning

This work assessed five ML algorithms (XGBoost, random forest or RF, SVM, Cat Boost, and logistic regression or LR) as well as two ensemble techniques, including voting (RF, XGBoost, CatBoost) and stacking (RF, XGBoost, CatBoost) classifiers, in order to train the cattle disease prediction model. This article used both Grid Search CV and Randomized Search CV to do hyper parameter optimization for each approach. K cross validation is performed. The value of K is 5. This work was able to completely assess a certain set of hyper parameter combinations thanks to this methodical technique, which made sure that every model had the optimal configuration. The hyper parameter settings for XGBoost model is displayed in Figure 5(a). The performance comparison among ML models with the Recursive Feature Addition feature selection approach is shown in Figure 5(b). All machine learning models performed much worse in the cattle disease prediction test when feature selection was not employed than when RFA was applied. XGBoost achieved highest accuracy value and F1 score value after RFA based feature selection.

The LR model obtained an overall accuracy 0.8780, according to Figure 5(b). The precision value, recall value, and F1-score values of the XGBoost model is highest, which was optimized using Recursive Feature Addition (RFA), were 0.9159, 0.9125, and 0.9125, respectively, while the total accuracy was 0.9129. The proposed XGBoost model's effectiveness is demonstrated by the confusion matrix figure in 5(c), which displays that most predictions are correctly classified along the diagonal for both healthy individuals and all eight sickness classifications. A small number of misclassifications do happen, mostly between clinically similar conditions (like FMD and LSD), although they are uncommon and have no effect on overall results. The predictions of our model were interpreted and explained in this work using SHapley Additive exPlanations (SHAP). The features impact on the models output and SHAP based summary plot are outlined in Figure 5(d). Each feature's contribution to the prediction of a particular illness class is represented by a different color in the bars. With the highest average SHAP values, the figure demonstrated that age and temperature are the most significant characteristics.

IV. RESULTS AND ANALYSIS

The proposed XGBoost-based cattle infectious illness prediction system performed best across all important measures, as Table 1 shows. The proposed XGBoost-based approach outperformed the results reported by previous works (i.e., Suparyati et al. [15], Bhardwaj et al. [12], and Naik et al. [17]) with accuracy value of 91.29%, precision value of 91.59%, recall value of 91.25%, and F1 value of 91.25%. The result comparison is performed on the same dataset and same number of classes. The accuracy gain and F1 score gain of the proposed XGBoost method were at least 1% and as much as 12% greater than those of previous works.

Because of its ability to manage tabular, heterogeneous features (numerical and categorical), support multi-class detection, and over fitting issue solving, the suggested XGBoost performs better for multi-class cattle infectious disease prediction. The SMOTE-NC oversampling technique was used to successfully handle class imbalance. RFA was used in the proposed strategy to choose the best features. The proposed XGBoost approach used both grid search and random search for hyper parameter tuning to enhance model performance.

V. CONCLUSION

This paper examined a machine learning based multi-class cattle infectious disease prediction method that considers one healthy class and seven main infectious bovine diseases. This paper gathered a comprehensive dataset on infectious diseases in cattle by considering the symptoms of the diseases and the validation of veterinary specialists. The SMOTENC technique is used in this paper to handle class imbalance issue. This paper employed hyper parameter tuning strategies based on random search and feature selection based on RFA method. Hyper parameter tuning strategies were used to train and improve five machine learning models, including ensemble approaches. With an accuracy value of 91.29 percent and precision value of 91.59 percent, the proposed XGBoost model performed better than other compared ML models for multi-class cattle infectious disease prediction. The results highlighted that, the proposed XGBoost model achieved at least 1% and up to 12% more accuracy value gain and F1 score gain than compared literary works. By using explainable AI techniques, especially SHAP, the system became more reliable and practical for veterinary professionals by offering transparent insights into model decisions. Future research challenges include, but are not limited to, block chain and AI-based cattle healthcare data transfer and monitoring systems, AI and LLM-based suggestions for treating cattle diseases, and IoT and DL-based real-time cattle disease monitoring.

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